

OS Project Plan

Created: May 1, 2026 **Status:** In Planning

Vision & Market Gap

Existing Linux distributions are primarily aimed at experienced users and developers. A clearly defined, youth-oriented alternative with a modern UI and AI integration does not currently exist on the market.

This project addresses three concrete problems:

Problem	Solution
Linux distros are visually outdated and complex	Custom desktop environment built on Wayland with a modern design system
File managers are based on outdated folder structures	AI-powered file manager with semantic search and automatic organization
Teenagers find no intuitive entry point into Linux	OS specifically designed for this target group, with onboarding and clear UX

Technological Advantage: By combining a local AI model in the file manager with a custom Wayland compositor, the result is a fully controllable, high-performance, and privacy-compliant platform.

Project Overview

A youth-friendly Linux-based operating system with modern aesthetics, an AI-powered file manager, and an intuitive user interface.

Technical Stack

Component	Decision
Kernel	Linux Kernel (adopted)
Bootloader	GRUB (custom styled)
Display System	Wayland (modern)

Package Manager	APT (Debian-based)
GUI Language	C/C++ (performance) + Python (tools & AI)
Languages	German + English

Design

Color Scheme

- **Dark Mode** – default, modern, easy on the eyes
- **Light Mode** – optional, switchable
- Smooth animations & transitions
- Target audience: teenagers

Design Inspiration

- Arc Browser (modern UI philosophy)
- Linear.app (clean aesthetics)
- Pop!_OS / elementaryOS (Linux distros with great design)

AI File Manager

The centerpiece of the OS — a completely new approach to file management.

Features

Smart Search

- Search by file content, not just file name
- Natural language queries: *"Show me the math document from last week"*
- Context-aware results

AI Auto-Organization

- AI automatically recognizes file contents
- Suggests meaningful folder structures
- Images sorted by subject (vacation, screenshots, etc.)
- Documents grouped by topic

-  **Predictive Access**
- AI learns the user's habits
 - Surfaces relevant files at the right time
 - Different suggestions in the morning vs. evening
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Learning Path & Milestones

Phase 1 - Foundations (current)

- ☐ Complete CS50 & obtain certificate
- ☐ Work through Nand2Tetris
- ☐ Install and explore Linux in VirtualBox
- ☐ Assembly basics (x86, NASM)

Phase 2 - First App

- ☐ Build file manager as a standalone Linux app
- ☐ Python + GTK or Electron as a base
- ☐ Implement AI search (locally, e.g. with SQLite + Embeddings)
- ☐ Publish on GitHub

Phase 3 - Desktop Environment

- ☐ Develop Wayland compositor (wlroots as base)
- ☐ Custom GUI components (taskbar, app launcher, etc.)
- ☐ Implement dark/light mode
- ☐ Animation system

Phase 4 - Custom Distro

- ☐ Use Debian as base
 - ☐ Integrate custom components
 - ☐ Style GRUB individually
 - ☐ Create ISO image (e.g. with Cubic)
 - ☐ Beta test with friends
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Recommended Resources

Resource	Purpose
osdev.org	OS development bible

"Writing a Simple OS from Scratch" (Nick Blundell)	OS entry point (free)
os-tutorial (GitHub)	Step-by-step OS building
wlroots Documentation	Wayland compositor
GTK Documentation	GUI with C/Python

This document was created for investors.